

CAC Hub — Installation Guide

Detailed step-by-step guide for installing the CAC Hub on a Raspberry Pi. The Hub is a central dashboard for controlling and monitoring multiple Pioneer CAC CD autochangers across the network.

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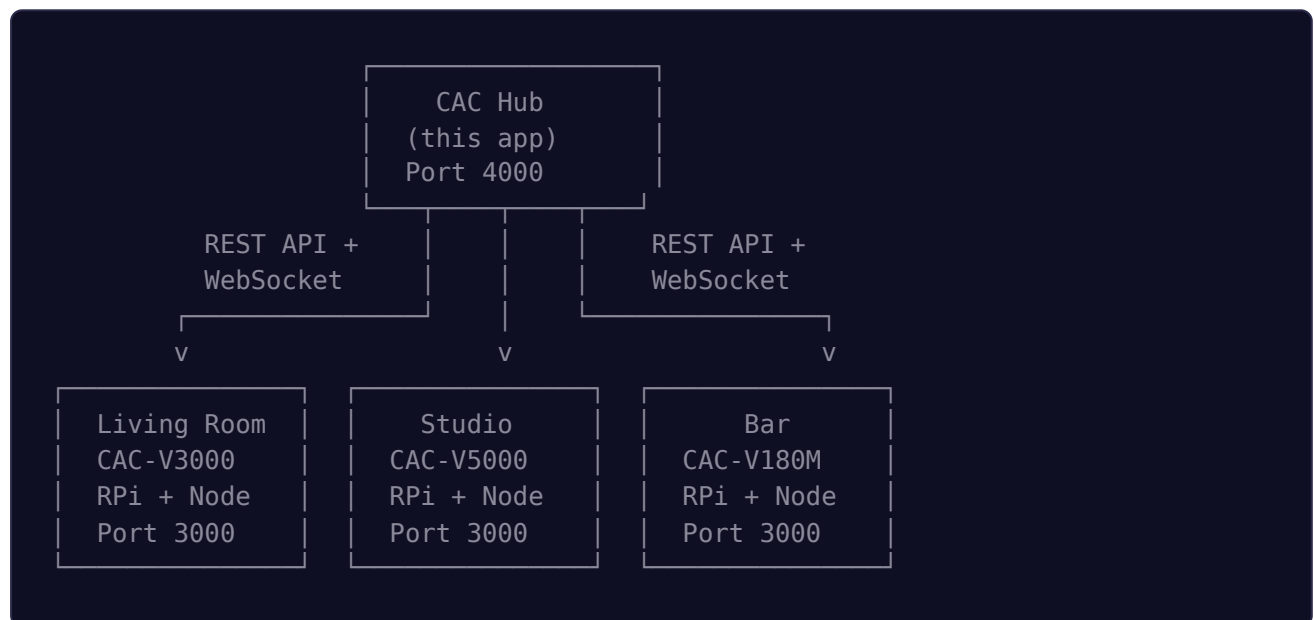
1. Overview

What is the CAC Hub?

The CAC Hub is a Node.js web application that connects via REST API and WebSocket to one or more CAC Controller nodes. It provides:

- — All changers at a glance with live status
- — Operate player, library, playlists, and scanner of each node
- — WebSocket events from all nodes are aggregated and displayed live
- — All node functions are accessible through the Hub with automatic API key authentication

Architecture



Important: Hub is Optional

Every node works fully on its own. The Hub is purely additive — it aggregates status and provides centralized control. If the Hub goes down, all nodes continue to operate unchanged.

2. Prerequisites

Hardware

- — Recommended: Pi 3B+, Pi 4, Pi 5, or Zero 2 W
 - The Hub requires serial port — only a network connection
 - Can run on the same Pi as a node (see Section 11)
 - Minimum 512 MB RAM
- — Minimum 8 GB
- — Appropriate USB power supply
- — WiFi or Ethernet

Software

- — Lite or Desktop (Debian Bookworm or newer)
- — Enabled
- — Version 18 or newer (20 LTS recommended)
- — For installation

Network Requirements

- All CAC Controller nodes must be reachable via HTTP from the Hub
- Default ports: Node 3000, Hub 4000
- Recommended: Static IP addresses for all Raspberry Pis

Node Requirements

- At least one configured CAC Controller Node
- Each node must have an generated (see Section 8)

3. Preparing the Raspberry Pi

If the Hub runs on the same Pi as a node that is already set up, this section can be skipped.

3.1 Installing the Operating System

1. Download : <https://www.raspberrypi.com/software/>
2. Flash Raspberry Pi OS Lite (64-bit) to the MicroSD card
3. In the Imager under :
 - Set hostname (e.g., `cac-hub`)
 - Enable SSH
 - Set username and password
 - Configure WiFi
 - Set timezone

3.2 Connect via SSH

```
ssh pi@<HUB_IP>
```

3.3 Update the System

```
sudo apt update && sudo apt upgrade -y
```

3.4 Configure Static IP Address (Recommended)

```
# Show current connection:
nmcli connection show

# Set static IP (example):
sudo nmcli connection modify "preconfigured" \
  ipv4.method manual \
  ipv4.addresses "192.168.1.200/24" \
  ipv4.gateway "192.168.1.1" \
  ipv4.dns "192.168.1.1"

# Restart connection:
sudo nmcli connection down "preconfigured"
sudo nmcli connection up "preconfigured"
```

3.5 Disable WiFi Power Saving (Recommended)

```
sudo tee /etc/NetworkManager/conf.d/wifi-powersave.conf > /dev/null << 'EOF'
[connection]
wifi.powersave = 2
```

```
EOF
sudo systemctl restart NetworkManager
```

4. Installing Node.js

If Node.js is already installed for a CAC Controller node, this step can be skipped.

4.1 Install Node.js 20 LTS

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
```

4.2 Verify Installation

```
node --version # Minimum v18.x
npm --version
```

5. Installing Build Tools

For the native SQLite extension (`better-sqlite3`):

```
sudo apt install -y build-essential python3
```

6. Installing the CAC Hub

Option A: From GitHub (Recommended)

```
cd /opt
sudo git clone https://github.com/NFDiJee/cac-controller-hub.git
sudo chown -R pi:pi /opt/cac-controller-hub
cd /opt/cac-controller-hub
npm install
```

Option B: Manual Copy

```
# On your local machine:
scp -r ./cac-controller-hub/* pi@<HUB_IP>:/opt/cac-controller-hub/

# On the Raspberry Pi:
cd /opt/cac-controller-hub
npm install
```

Verify Installation

```
ls -la /opt/cac-controller-hub/
# Should contain: server.js, package.json, src/, public/, ...

ls -la /opt/cac-controller-hub/node_modules/
# Should show installed packages
```

7. First Start and Testing

7.1 Start the Server Manually

```
cd /opt/cac-controller-hub
node server.js
```

Expected output:

```
[Hub] Initializing database...
[NodeManager] Connecting to 0 nodes...
[Hub] CAC Hub running on http://0.0.0.0:4000
```

7.2 Open in Browser

```
http://<HUB_IP>:4000
```

You should see the Hub dashboard — still empty since no nodes are configured. The dashboard shows:

- Empty node card view
- Navigation: Dashboard, Settings
- Language toggle (DE/EN)

7.3 Stop the Server

Press `Ctrl+C` in the terminal.

8. Adding Nodes

8.1 Generate API Key on the Node

Before a node can be registered in the Hub, it must have an API key:

1. Open the node's web interface: `http://<NODE_IP>:3000`
2. Under `Settings`, open the `API Keys` section
3. Enter a `Room` (e.g., "Living Room")
4. Enter a `Floor` (e.g., "Ground Floor")
5. Click `Generate`
6. Copy the generated key (needed in the Hub)
7. Click `Close`

8.2 Add Node in the Hub

1. Open the Hub in browser: `http://<HUB_IP>:4000`
2. Switch to `Nodes`

3. In the section:

- : `http://<NODE_IP>:3000` (full URL with port)
- : Paste the copied API key

4. Click

- On success: , , and are auto-filled
- On failure: Check URL and key, test network connectivity

5. Click

8.3 Verify Connection

After adding:

- The node should appear on the dashboard as (green dot)
- Player status cards should show the current state of the node
- Quick controls (Play, Pause, Stop, Next) should work

8.4 Test Connection via Command Line

If the Hub UI doesn't work, test the connection manually:

```
# From the Hub Pi:
curl -H "X-API-Key: YOUR_API_KEY" http://<NODE_IP>:3000/api/node/info
```

Expected response:

```
{
  "name": "Living Room",
  "room": "Ground Floor",
  "model": "CAC-V3000",
  "version": "1.0.0"
}
```

If no response: check network connectivity, firewall rules, node service status.

9. Setting Up the Systemd Service

9.1 Copy Service File

```
sudo cp /opt/cac-controller-hub/cac-hub.service /etc/systemd/system/
```

9.2 Adjust Service File (If Needed)

```
sudo nano /etc/systemd/system/cac-hub.service
```

Contents:

```
[Unit]
Description=CAC Hub - Central Dashboard for Pioneer CD Autochanger Nodes
After=network.target

[Service]
Type=simple
User=pi
WorkingDirectory=/opt/cac-controller-hub
ExecStart=/usr/bin/node server.js
Restart=always
RestartSec=5
Environment=NODE_ENV=production

[Install]
WantedBy=multi-user.target
```

Change `User=` to the correct user if not `pi`.

9.3 Enable and Start the Service

```
sudo systemctl daemon-reload
sudo systemctl enable cac-hub
sudo systemctl start cac-hub
```

9.4 Check Status

```
sudo systemctl status cac-hub
```

9.5 View Logs

```
# Live logs:
sudo journalctl -u cac-hub -f

# Last 50 lines:
sudo journalctl -u cac-hub --no-pager -n 50
```

9.6 Test Autostart

```
sudo reboot
```

After reboot: `http://<HUB_IP>:4000` should be accessible and all nodes should connect automatically.

10. Configuring Hub Settings

In the browser under (on the Hub):

10.1 General

Setting	Default	Description
Hub Name	CAC Hub	Display name in browser tab and header
Language	auto	Automatic (by browser language), German, or English

10.2 Network

Setting	Default	Description
Port	4000	HTTP server port (restart service after change)

10.3 Connection

Setting	Default	Description
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Reconnect Interval	10000 ms	Wait time between reconnection attempts
Health Check Interval	30000 ms	Interval for periodic status polling

10.4 Apply Changes

Port changes require a service restart:

```
sudo systemctl restart cac-hub
```

Language and name changes take effect immediately.

11. Hub and Node on the Same Pi

The Hub can easily run on the same Raspberry Pi as a CAC Controller node. This is the simplest configuration for small setups.

Port Configuration

- : Port 3000 (default)
- : Port 4000 (default)
- Both run in parallel without conflicts

Set Up Both Services

```
# If not already done -- Node service:
sudo systemctl enable cac-controller
sudo systemctl start cac-controller

# Hub service:
sudo systemctl enable cac-hub
sudo systemctl start cac-hub
```

Connect Node via localhost

Add the node in the Hub with:

- : `http://localhost:3000`
- : The key generated on the node

Resource Usage

Both applications together require:

- Approx. 80–120 MB RAM
- Minimal CPU load (Raspberry Pi 3 or newer is sufficient)

12. Setting Up Multiple Nodes

12.1 Network Planning

For a multi-node setup, consider:

Device	IP Address	Port	Hostname
Hub	192.168.1.200	4000	cac-hub
Node Living Room	192.168.1.101	3000	cac-livingroom
Node Studio	192.168.1.102	3000	cac-studio
Node Bar	192.168.1.103	3000	cac-bar

12.2 Order of Setup

1. Install and configure all nodes (see CAC Controller Installation Guide)
2. Generate an API key on each node
3. Install the Hub
4. Add all nodes in the Hub

12.3 Ordering

Nodes are displayed on the dashboard in the order they were added. The order can be adjusted via the `sort_order` field in the database.

13. Updates

From GitHub Repository

```
cd /opt/cac-controller-hub
git pull
npm install
sudo systemctl restart cac-hub
```

Manual Update

1. Copy new files to the Pi
2. Run `npm install`
3. Restart service: `sudo systemctl restart cac-hub`

After an Update

- All node connections are automatically restored
- The Hub database (node configurations) is preserved
- Settings are preserved

14. Troubleshooting

14.1 Hub Does Not Start

```
# Check logs:
sudo journalctl -u cac-hub --no-pager -n 50

# Start manually for details:
cd /opt/cac-controller-hub && node server.js
```

- Port 4000 in use: `sudo lsof -i :4000`
- Missing node_modules: run `npm install` again
- Wrong user in service: check `User=` in the service file

14.2 Node Not Reachable

```
# Test from Hub Pi:
curl -H "X-API-Key: YOUR_KEY" http://<NODE_IP>:3000/api/node/info

# Ping test:
ping <NODE_IP>

# Port test:
nc -zv <NODE_IP> 3000
```

- Node service not running: check `sudo systemctl status cac-controller` on the node
- Wrong API key
- Firewall blocking port 3000
- WiFi power saving on node or Hub

14.3 Node Shows "Offline" on Dashboard

- The Hub automatically retries connection every 10 seconds
- Check URL and API key in Hub, check node service
- Check network connectivity

14.4 WebSocket Connection Drops

WebSocket connections can be interrupted by network timeouts. The Hub has three safeguards:

1. — Automatically reconnects (every 10s)
2. — REST API polling as backup (every 30s)
3. — Browser client reconnects automatically

14.5 Port 4000 in Use

```
# Find the process:
sudo lsof -i :4000

# Choose a different port in Hub settings
# Then restart service:
sudo systemctl restart cac-hub
```

14.6 Proxy Error (502 Bad Gateway)

When Hub commands to nodes fail:

```
# Test directly at the node (without Hub):
curl http://<NODE_IP>:3000/api/player/1/status

# Test with API key:
curl -H "X-API-Key: KEY" http://<NODE_IP>:3000/api/player/1/status
```

Node is unreachable or the API key is incorrect.

14.7 Cover Images Not Displayed

Cover images are loaded via the Hub proxy. If they are missing:

1. Check directly on the node: `http://<NODE_IP>:3000` — are covers visible there?
2. Test proxy: `http://<HUB_IP>:4000/api/nodes/1/cover/cover_1.jpg`

15. Uninstallation

Remove Service

```
sudo systemctl stop cac-hub
sudo systemctl disable cac-hub
sudo rm /etc/systemd/system/cac-hub.service
sudo systemctl daemon-reload
```

Delete Files

```
sudo rm -rf /opt/cac-controller-hub
```

Remove Node.js (Only if no Node Runs on This Pi)

```
sudo apt purge -y nodejs
```